

International Finance

Lecture III – March 3rd

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IES FSV UK – Spring 2026

Moodle: <https://dl1.cuni.cz/course/view.php?id=4605>

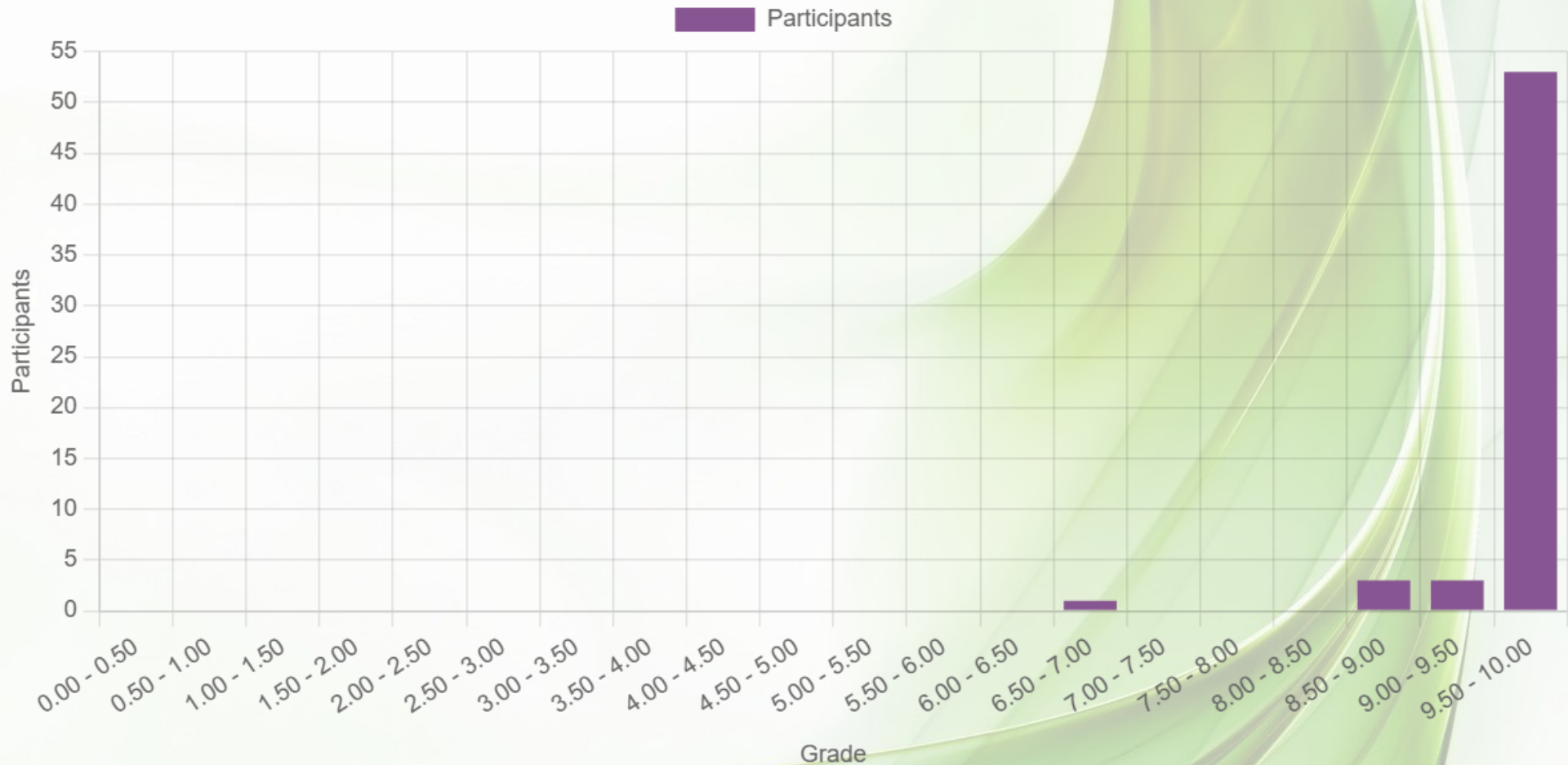
(International Finance JEB050/JPB335)

Course enrolment key: **PostUSD?2026**

Outline

- Course organization
 - Attendance, quiz
 - Quiz #3: will be open by tomorrow (Wednesday) midnight
 - The usual deadline: next Monday midnight
- Balance of payments and international investment position
 - Stocks vs. flows
- Balance of payments: accounting
- External imbalances: macroeconomic identities and national accounting

Quiz #2



60 participants/64 attempts

Reading for This Week:

- Additional materials on money, multiplication process – for those of you without prior knowledge of monetary economics
- Cipriani, Marco, Linda S. Goldberg, and Gabriele La Spada. 2023. "Financial Sanctions, SWIFT, and the Architecture of the International Payment System."
 - Can you briefly outline the difference between intra-country payments and cross-border payments? What is “correspondent banking”? What is the role of SWIFT in cross-border transactions?
 - Are there any competitors to SWIFT?
 - What is the meaning of the following acronyms: OFAC, AML, SWIFT?
 - How common are sanctions that restrict access of target countries to the infrastructures and institutions that execute international payments?

Reading for the Next Week:

- Krugman & Obstfeld & Melitz: International Economics: chapter 13
- Bellocchi & Travaglini (2025): The Trade Deficit Delusion: Why Tariffs Will Not Make America Great Again
- Additional reading:
 - Yang (2012): Aggregate Savings and External Imbalances in China. AER
 - Boileau & Normandin (2012): Do tax cuts generate twin deficits? A multi-country analysis. The Canadian Journal of Economics / Revue canadienne d'Economie, Vol. 45, No.4
 - Pages 1667-1671 are sufficient at this stage

Course Outline (Changes Possible)

1. Introduction to international finance. Foreign exchange market: main actors, their role, types of interactions. Brief outline of the history of international finance. (February 17th)
2. International transfers/payments. Channels, related instruments. (February 24th)
3. **Balance of payments structure and national income accounting for an open economy. International investment position** (March 3rd)
4. Introduction to exchange rate determination: asset approach. Covered and uncovered interest rate parity. (March 10th)
5. Money, interest rates and exchange rates. Long-run aspects of exchange rate determination. Purchasing power parity. Overshooting. (March 17th)
6. Prices and exchange rates continued: Fischer effect. Exchange rates and competitiveness: real (effective) exchange rates, Balassa-Samuelson effect. Application on the nominal and real convergence in the EU (March 24th)
7. Introduction to the AA-DD model. Temporary and permanent changes in fiscal and monetary policies. (March 31st)
8. The AA-DD model: XX curve. Application on policy analysis. (April 7th)
9. Policy options for reaching internal and external balance. Swan diagram. Fixed exchange rates: macroeconomic implications. Interventions and sterilisation. Policy trilemma. (April 14th)
10. **Monetary integration: costs and benefits. Monetary integration: effects – empirical research.** (April 21st)
11. A brief introduction to the models of balance of payments crises. (April 28th)
12. Debts and (developing) countries. **Deadline for the team assignment/paper** (May 5th)
13. The future of IMS, role of the USD. Discussion of current topics. (May 12th)
14. **Reserved for a make-up class (possibly for April 21st). (May 19th)**

National Accounts and the Balance of Payments

The background of the slide features abstract, flowing, wavy lines in various shades of green and white, creating a sense of movement and depth. The lines are more prominent on the right side of the slide, curving upwards and outwards.

Data on International Transactions

- **Balance of Payments**

- A systematic record of all economic transactions between residents and non-residents during a given period (year, quarter)

- **Flow** indicator!

- **International Investment Position**

- A record of all external assets and external liabilities of the economy
- i.e. IIP records all assets held by our residents in foreign countries and by foreigners in our country

- **Stock** indicator!

- E.g. compare the numbers for “reserves” in the balance of payments and investment position

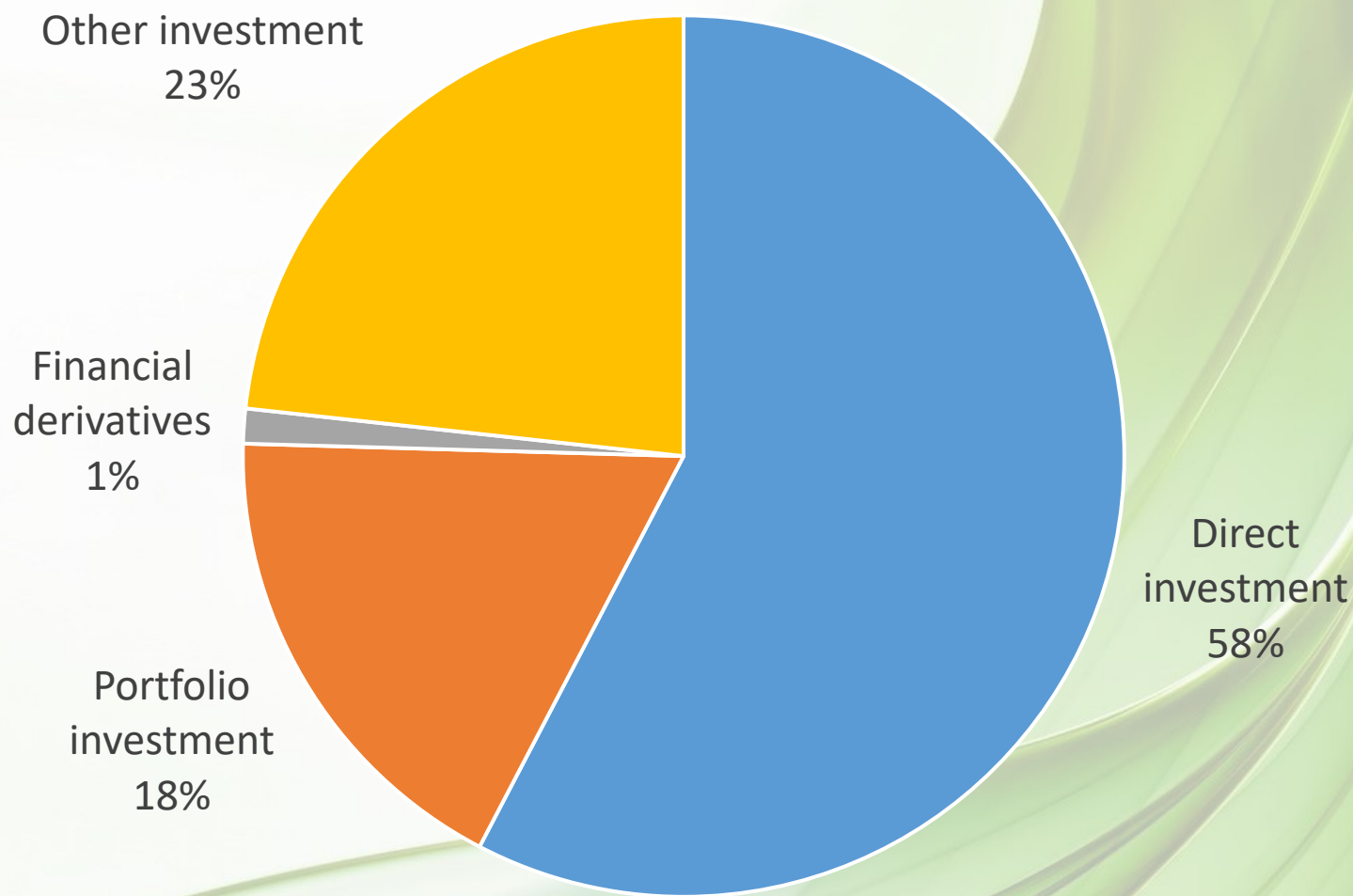
Czech Investment Position

- The **investment position of the Czech Republic vis-a-vis non-residents**
 - *A survey of the positions of financial assets and liabilities of all sectors of the domestic economy (the Government sector, the banking sector incl. the CNB and the corporate sector) in relation to the non-residents of the Czech Republic as at a certain date (as at the last day of the reporting period).*
- The data of the investment position are monitored on the **gross-principle**, i.e. financial assets and liabilities for the individual items are monitored separately.
- The difference between financial assets and liabilities represents the resulting **balance of the investment position**.

CR: Investment Position, September 2025, mil. CZK

ASSETS	10 034 715.5		LIABILITIES	10 842 845.9
Direct investment abroad	2 608 050.5		Direct investment in the Czech Republic	6 252 180.6
Portfolio investment	1 576 772.5		Portfolio investment	1 928 872.3
Equity securities	1 111 068.4		Equity securities	409 261.1
Debt securities	465 704.1		Debt securities	1 519 611.1
Financial derivatives	150 343.4		Financial derivatives	138 474.8
Other investment	2 158 485.1		Other investment	2 523 318.2
Reserves	3 541 063.9			
Monetary gold	170 326.2			
Special drawing rights (SDR)	73 007.1			
Reserve position in the IMF	16 049.9			
Other reserve assets	3 281 680.7			
NET INVESTMENT POSITION: -808 130.4 mil. CZK				

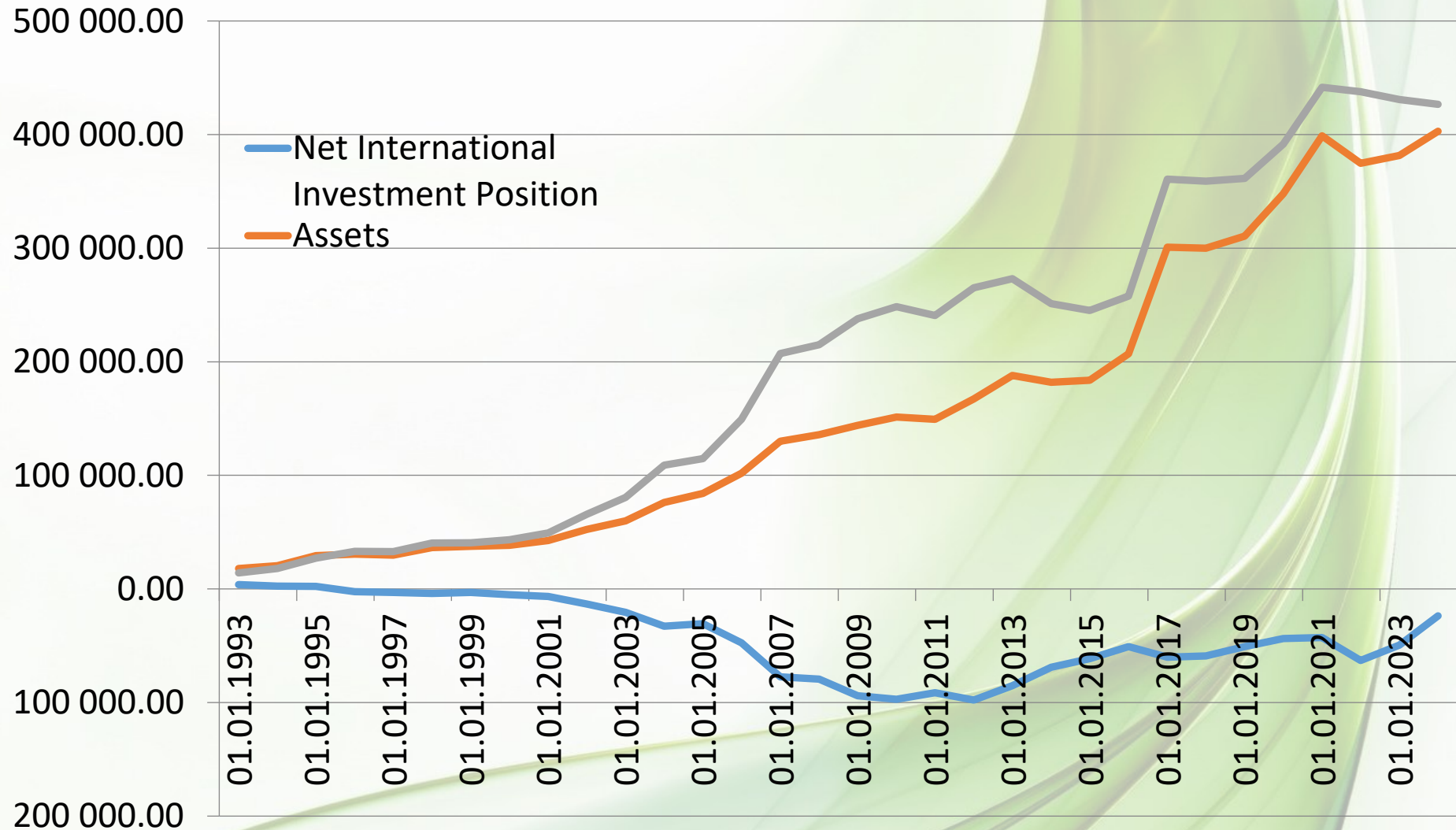
CR: IP, Structure of Liabilities, Sept 30, 2025



Source: CNB

CR: Investment Position 1993-2025

Millions of USD



Source of data: Czech National Bank, ARAD

Summary: What Can you Find in the Investment Position?

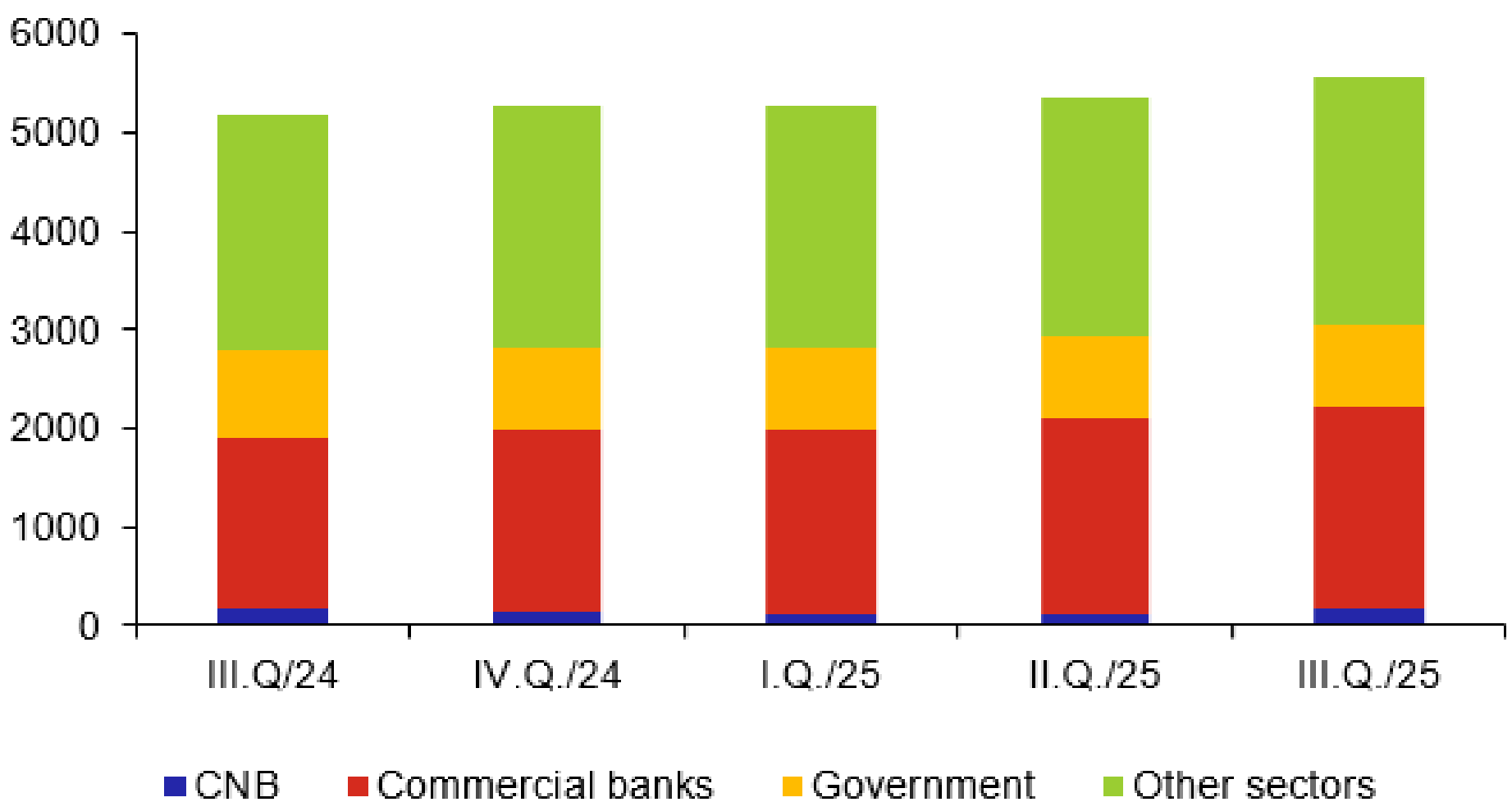
- Data on **stocks**

- **Stocks** of foreign exchange reserves
- Total **stock** of FDI in the country (and total stock of our direct investment abroad)
- **Stock of** foreign debt of a country
 - Czech Republic: 4 546 070.6 mil. CZK or 197 939,2 mil. USD (Sept 30, 2023)

Foreign Debt / Foreign Indebtedness

- **Foreign debt $\neq$ public debt**
 - Could there be any exceptions?
- **Foreign indebtedness of the Czech Republic** is defined as a **survey of financial liabilities** of the sectors of the domestic economy vis-a-vis nonresidents of the CR which are of debt character
 - i.e. liabilities with contractually determined maturity for which a yield in the form of interest belongs to the creditor.
- **Foreign indebtedness does not include** the positions of investments into equity securities, i.e. direct investments – owners' capital, portfolio investments - equity securities and ownership interests.

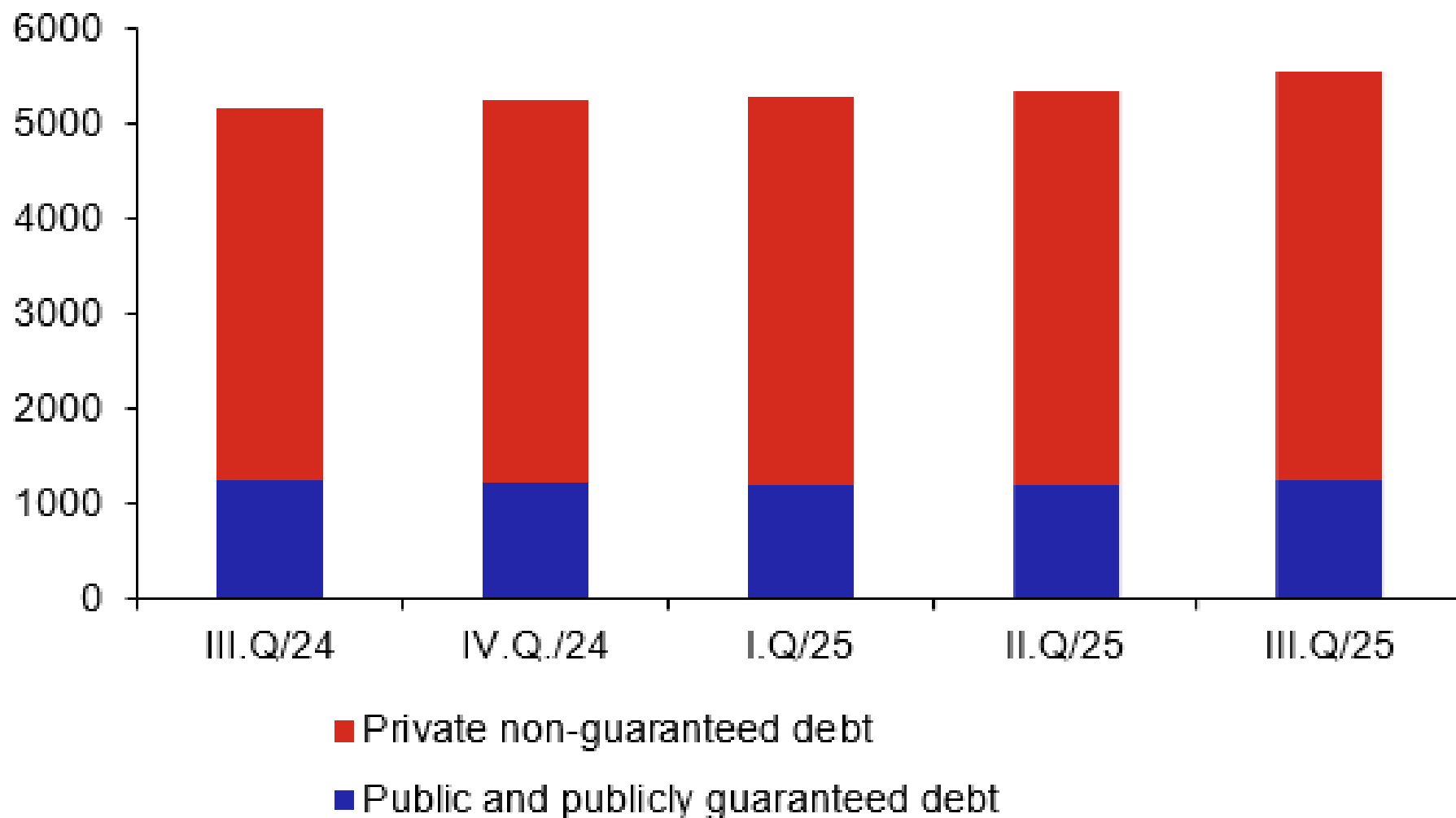
Czech External Debt by Debtor (CZK bil.)



Source:

https://www.cnb.cz/en/statistics/bop_stat/investment_position/the-czech-republics-international-investment-position-and-external-debt/

Czech External Debt (CZK bil.)



Source:

https://www.cnb.cz/en/statistics/bop_stat/investment_position/the-czech-republics-international-investment-position-and-external-debt/

If I need flows?

Balance of Payments

Structure of Balance of Payments

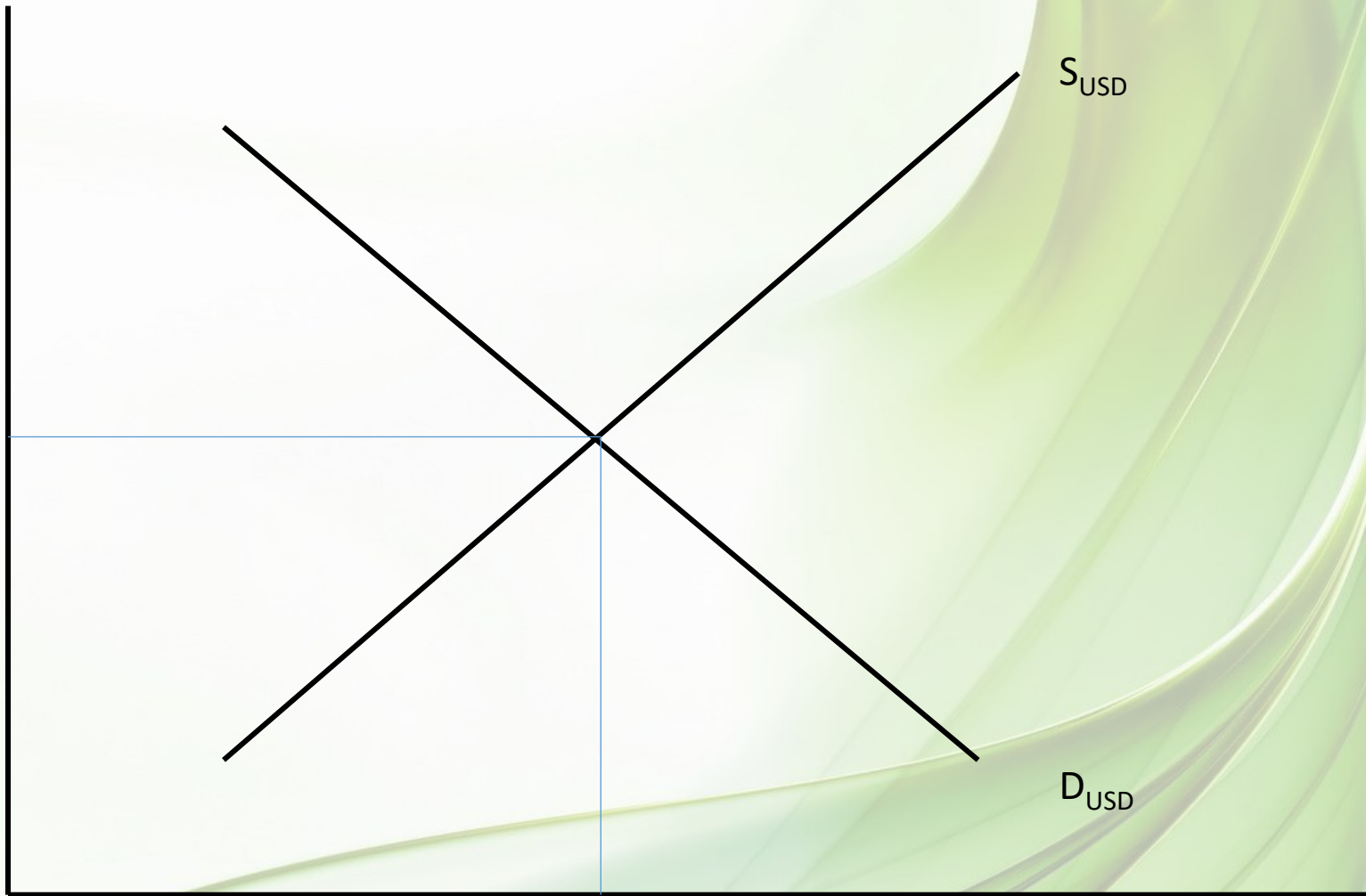
	Resources ↑	Resources ↓
	Credit (+)	Debit (-)
Current Account	Export of goods	Import of goods
	Export of services	Import of services
	Primary income inflow	Primary income outflow
	Secondary income inflow	Secondary income outflow
Capital Account	Capital transfers inflow	Capital transfers outflow
Financial Account	Decrease of external claims	Increase of external claims
	Increase of external liabilities	Decrease of external liabilities
Errors and Omissions		
Reserves	Decrease of reserves	Increase of reserves

BP Accounting

- Credit items
 - Reflect transactions that give rise to payments inward to the country
- Debit items
 - Reflect transactions that give rise to payments outward from the country

Foreign Exchange Market

Exch rate
CZK/USD



Dollars traded

Current Account and Components

- *The current account shows flows of goods, services, primary income, and secondary income between residents and nonresidents.*

Current account

Goods and services

Goods

Services

Primary income

Compensation of employees

Interest

Distributed income of corporations

Reinvested earnings

Rent

Secondary income

Current taxes on income, wealth, etc.

Net nonlife insurance premiums

Nonlife insurance claims

Current international cooperation

Miscellaneous current transfers

Adjustment for change in pension entitlements

Capital Account

- The capital account shows credit and debit entries for **non-produced nonfinancial assets** and **capital transfers** between residents and nonresidents.
 - Non-produced, nonfinancial assets consist of:
 - natural resources;
 - contracts, leases, and licenses;
 - marketing assets (and goodwill).
 - brand names, mastheads, trademarks, logos, and domain names.
 - Capital transfers
 - Debt forgiveness
 - Investment grants
 - One-off guarantees
- CA records **acquisitions and disposals of non-produced nonfinancial assets**, such as land sold to embassies and sales of leases and licenses, as well as **capital transfers**, that is, the provision of resources for capital purposes by one party without anything of economic value being supplied as a direct return to that party.

Table 13.1. Overview of the Capital Account

	Credits	Debits
Current account balance		
Acquisitions (DR.)/disposals (CR.) of nonproduced, nonfinancial assets		
Natural resources		
Contracts, leases, and licenses		
Marketing assets		
Capital transfers		
Debt forgiveness		
Other		
Capital account balance		
Net lending (+)/net borrowing (–) (from current and capital accounts)		

Note: This table is expository; for Standard Components, see Appendix 9.

Financial Account

- Records flows of capital

Financial Account	Credit	Debit
Direct investment	FDI	
Portfolio investment		
Other investment – long term		
Other investment – short term		

Direct Investment: Definition

- Investment from one country into another (normally by companies rather than governments) that involves establishing operations or acquiring tangible assets, including stakes in other businesses.
- The purchase or establishment of income-generating assets in a foreign country that entails the control of the operation or organisation.
- FDI is distinguished from portfolio foreign investment (the purchase of one country's securities by nationals of another country) **by the element of control.**
- Standard definitions of control use **the internationally agreed 10 per cent threshold** of voting shares

BP Accounting (2)

- **“Double-entry bookkeeping”**
 - Every transaction has two faces (credit and debit) and it is recorded twice

Balance of Payments

Item	Credit	Debit	Macrostructure
Trade in goods	+100		Current Account
Services			
Primary income			
Secondary income			
Capital account			Capital Account
Direct inv.			Financial Account
Portfolio inv.			
Other capital		-100	
Errors & omissions			
Change in reserves			Reserves

Main Balances for Analytical Purposes

- We can calculate balances for each item of the BP
- However, some are more interesting and more often used than others:
 - **Balance of trade:** export – import of **merchandise**
 - Balance of trade in goods and services
 - **Current account**
 - Basic balance = CA + long-term capital
 - **Overall balance** or Official settlements balance

BP and Double Entry Bookkeeping: Implications

Current Account

+

Capital Account

+

Financial Account

+

Change in Reserves

= 0

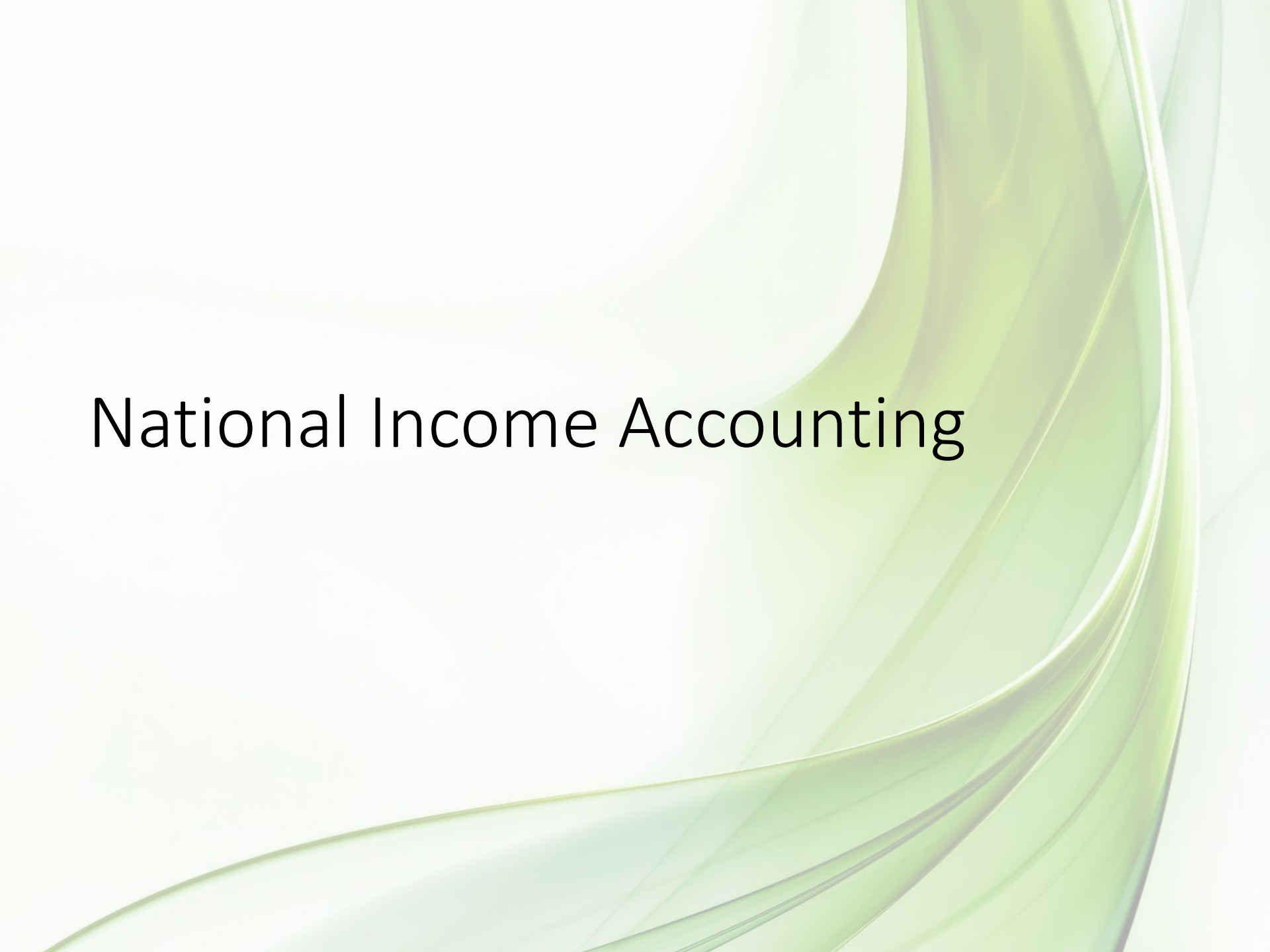
Analysis of BP

- Simplified structure: CA, FA, reserves
- We are interested in the structure of BP
 - Surplus or deficit in goods and services?
 - Surplus or deficit in current account?
 - Are stable components of financial account sufficient for financing of CA deficit?
- ... and in sustainability of recent development
 - Reasonable BT deficit to GDP or CA deficit to GDP ratio
 - Reasonable debt indicators

BP Accounting

- Notice that the upper part of the BP includes balances of what can be described as “real flows” and the lower part “financing”
- Surplus in the “upper part” leads to accumulation of assets (or reduction of debts)
 - BP and Investment Position are connected!
- Deficit in the upper part leads to sale of “our” assets (or accumulation of debts)

National Income Accounting

The background of the slide features a series of smooth, flowing, wavy lines in shades of light green and pale yellow. These lines originate from the bottom left and curve upwards and to the right, creating a sense of movement and depth. The overall aesthetic is clean and modern.

National Income Accounting

- For basic concepts (GNP, GDP, C, I, X, M) – please see KOM (chapter 13)

Basic Identities

- The basic macroeconomic identity:

$$Y = C + I + G + (X - M)$$

- can be rearranged:

$$(X - M) = Y - (C + I + G) = Y - A$$

- i.e. CA is the difference between output and domestic **absorption (total final use in our economy)**
- Interpretation:
 - CA happens when we consume or invest more than our economy produces

Role of Savings

- S national saving (total: public + private):

$$S = Y - C - G$$

- Closed economy:

$$S + C + G = C + I + G$$

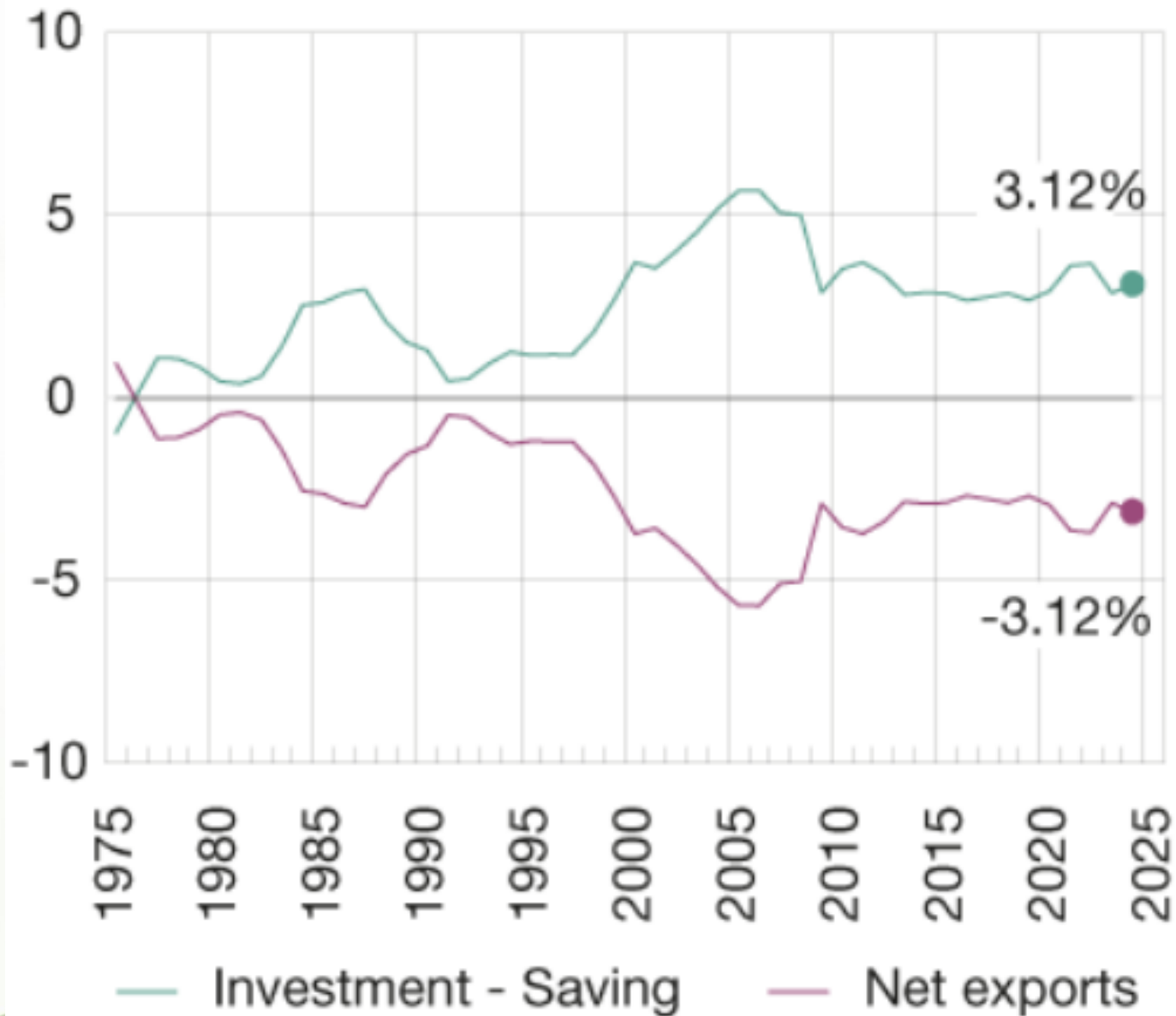
$$S = I$$

- Open economy: $S + C + G = C + I + G + (X - M)$

$$S = I + (X - M)$$

- Open economy can save either by building its capital stock or **by acquiring foreign wealth**
- Open economy **can invest more than domestic savings** allow if it incurs foreign debt or sells foreign assets!

a. The saving-investment gap (% of GDP)



Source: Bellocchi & Travaglini (2025): The Trade Deficit Delusion

Interpretation

- Current account shows us “national savings”
- When current account positive, the country is “saving” assets for future (or repaying old debts)
- National savings are composed of private and government savings
- Crucial factor: **intertemporal role of CA deficits/surpluses**
 - Intertemporal → we may need to model how households decide about current and future consumption

Private and Government Saving

- Private saving: $S^P = Y - T - C$

- Government saving: $S^G = T - G$

- Total national savings:

$$S = Y - C - G = (Y - T - C) + (T - G) = S^P + S^G$$

- Therefore if $S = I + CA$, we can further infer that:

$$S^P = I + CA - S^G = I + CA - (T - G) = I + CA + (G - T)$$

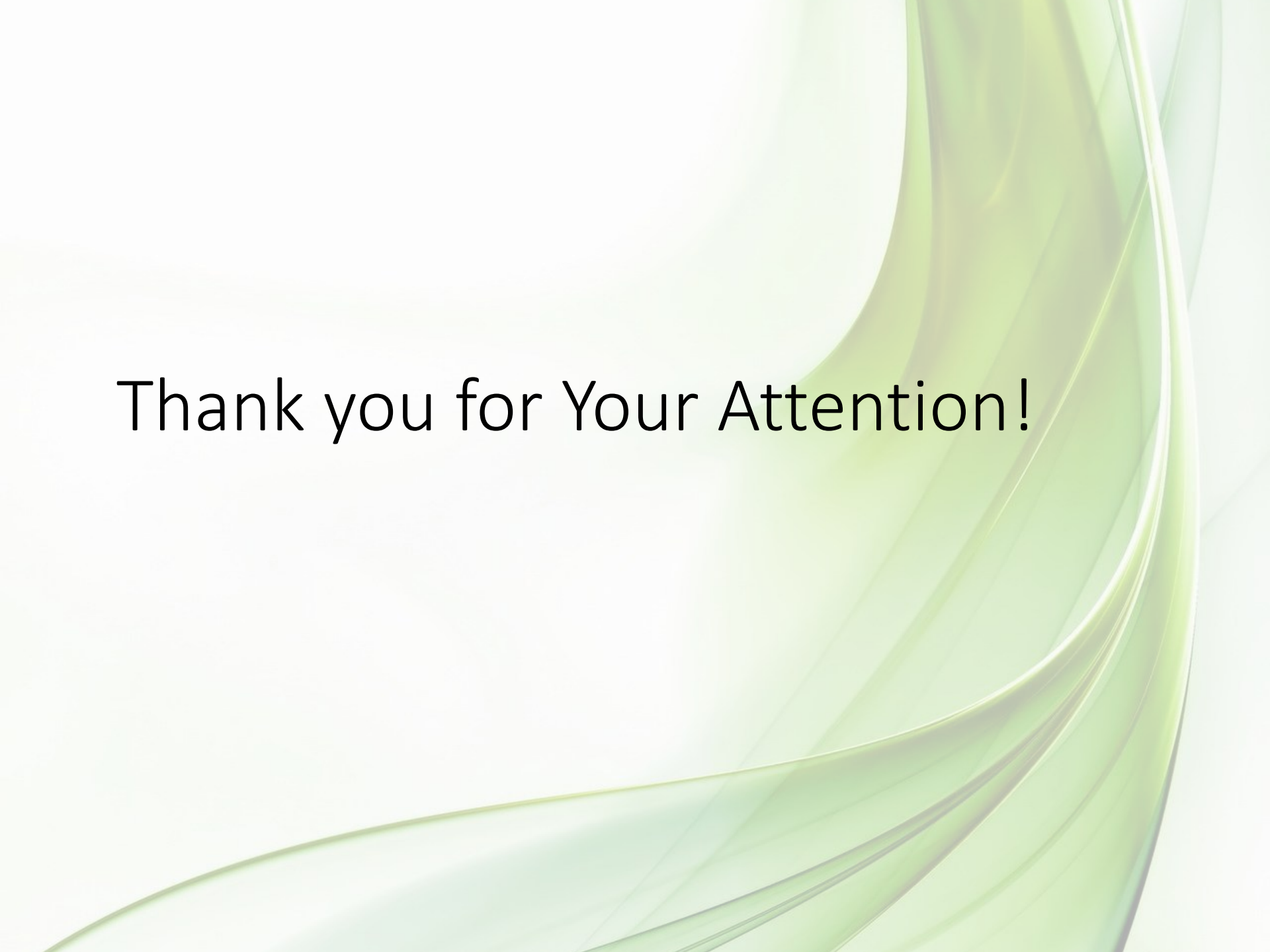
or:

$$CA = (S^P - I) + (T - G)$$

The Original “Twin Deficit”

USA (% GDP)				
Year	NX	S	I	T-G
1981	0.3	18.0	16.9	-1.0
1982	0.0	17.6	14.1	-3.5
1983	-1.0	17.7	14.8	-3.8
1984	-2.4	18.4	17.9	-2.9
1985	-2.9	17.5	16.8	-3.5

Source: Krugman - Obstfeld: International Economics

The background of the slide features abstract, flowing, and layered green and white lines that create a sense of movement and depth, primarily concentrated on the right side of the frame.

Thank you for Your Attention!

References

- Bellocchi & Travaglini (2025): The Trade Deficit Delusion: Why Tariffs Will Not Make America Great Again
- Boileau & Normandin (2012): Do tax cuts generate twin deficits? A multi-country analysis. The Canadian Journal of Economics / Revue canadienne d'Economie, Vol. 45, No.4
- CNB: International Investment Position
- Krugman & Obstfeld & Melitz: International Economics (chapter 13)
- IMF: Sixth Edition of the IMF's Balance of Payments and International Investment Position Manual (BPM6)
- World Development Indicators
- S&P 2018: Twin deficits: mind the gap(s)
- Yang (2012): Aggregate Savings and External Imbalances in China. AER